Take-Home Assignment: REFIT Data Cleaning and Appliance Disaggregation

Scenario

Flock Energy works with electricity distribution utilities and smart-meter data. The goal is to understand household electricity use and identify opportunities to reduce consumption. A recurring challenge is estimating appliance usage from whole-home measurements when appliances are not directly metered. Build a manageable, reproducible pipeline to clean raw REFIT data and develop an appliance-disaggregation model using the official cleaned data.

Time and scope: Aim for 4-6 hours. We care more about thoughtful choices, clear reasoning, and a reproducible analysis than completing every possible analysis. If you run out of time, briefly explain what you would do next and why.

Dataset

Use REFIT Electrical Load Measurements from the University of Strathclyde. It covers 20 UK households and provides aggregate and appliance-level power readings. Consult the dataset documentation and state the households/files used, date coverage, sampling interval, units, and relevant limitations.

- Raw REFIT data (.7z): https://pureportal.strath.ac.uk/files/62090177/REFIT_RAW_081116.7z
- Cleaned REFIT data (.7z): https://pureportal.strath.ac.uk/files/62090184/CLEAN_REFIT_081116.7z
- Cleaned REFIT readme: pure.strath.ac.uk/ws/portalfiles/portal/62090183/CLEAN_READ_ME_081116.txt
- Dataset page and documentation: https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements/

Requirements

1. Basic inspection and cleaning of raw data

Use raw data for this section and select House 1. Resample its readings to 1-minute intervals. This task focuses on essential data-handling skills; you are not expected to reproduce every specialised correction in the official REFIT cleaning pipeline.

- Check timestamp parsing and ordering, duplicate timestamps, missing readings, irregular intervals, and obvious impossible or extreme values. Summarise the findings.
- Create a simple, documented pipeline: sort readings, remove duplicates using a stated rule, and handle short missing gaps. Flag longer outages rather than filling them without justification.
- Plot aggregate power and at least two appliance channels before and after cleaning.

2. Exploratory analysis of washing-machine use

Use washing-machine appliance readings from the official cleaned REFIT data across households where this channel is available. Analyse at least 4 households at 1-minute resolution with a washing machine channel (more is welcome if time permits). State which homes and periods you used.

- Summarise availability and data quality: date range, missing intervals, zero/off time, and number of detected operating events or cycles. State the threshold or rule used to identify an active cycle.
- Define a simple and reasonable rule for identifying an active washing-machine cycle. Explain the threshold/rule in plain language.

- Describe usage patterns by hour of day and day of week, and compare at least one cycle characteristic (for example, duration, peak power, or energy per cycle) across selected homes.
- Include: (i) one representative 24-hour aggregate + washing-machine trace, (ii) one or more detected washing-machine cycles, and (iii) one cross-house comparison. Briefly interpret each plot.

3. Build and evaluate an appliance-disaggregation model

Use the official cleaned REFIT data. Estimate washing-machine power/load from the corresponding household aggregate signal at 1-minute resolution. State the homes and periods included.

- Use any model that you think is appropriate for this case. Examples include random forest/gradient boosting, a basic neural network, Seq2Point, or another approach you can explain. Model complexity is not part of the assessment.
- Include at least one simple baseline (for example, always-off or a simple usage-profile baseline) and compare your model against it.
- Use a sensible train/test or train/validation/test split. Prefer splitting by household or by time rather than randomly mixing overlapping time windows. Explain your choice and how it avoids data leakage.
- Choose appropriate metrics and explain why they are useful. Include one clear actual-vs-predicted plot on evaluation data.
- Briefly describe one or two failure cases or situations where the model is likely to struggle.

4. Practical implications

- Identify one practical energy-saving or customer-insight opportunity suggested by your analysis. Keep it realistic and explain how you would measure whether it worked.
- Briefly discuss limitations such as overlapping appliance signatures, missing data, household differences, and applying results from UK homes to India.
- In a short paragraph, explain what would become harder if only 15-minute or 30-minute smart-meter data were available.

Research guidance

You may consult research articles that use REFIT to identify suitable NILM algorithms, preprocessing choices, and evaluation methods. Cite the articles used, explain which ideas you adopted, and implement and evaluate your own work.

Deliverables

Submit a public GitHub repository containing the following:

File	Content
Source code	Notebook(s) or scripts that run end-to-end from the supplied data to cleaning, EDA, model predictions, and evaluation.
README.md	What you built, setup and run instructions, assumptions, preprocessing decisions, validation design, trade-offs, AI tools used, and possible improvements.
DATA.md	Raw-data coverage and quality findings, cleaning steps, decisions about gaps/outliers, and remaining concerns.
RESULTS.md	Washing-machine EDA, plots, model results, validation approach, failure analysis, and interpretation.
Recommendation.md	One-page utility-facing summary: suitable uses and limits of the model. Also include the answers to Section 4 questions

What we are assessing: correct data handling, sensible reasoning, modelling and validation, reproducibility, and the ability to explain your choices clearly. A simple, correct solution will score better than a complex solution that is difficult to justify or validate.